

Ontology of Becoming

Argument summary

Primitive. A *law* is a map m on a carrier. $\text{Fix}m$ means $\exists x. x = m x$. $\text{Run}m$ means: the recursion $x_{n+1} = m x_n$ is solved from every seed, and consecutive terms differ.

Df. m is *structureless* if it commutes with every value-relabeling of its carrier. On two values this is commutation with the exchange. On $\mathbb{B}$ this coincides with definability from equality (theorem).

L1 (Arena). A carrier with decidable equality admits a non-identity structureless law iff it has exactly two elements. (One element: only the identity. Three or more pairwise-distinct elements: every equivariant law is the identity. A moved point and its image exhaust the carrier. On two elements the exchange moves both.) The same equivalence holds for mark-respecting equivariance: a law commuting with transpositions that fix a marked value moves the mark, or moves every value, iff the carrier has two elements.

L2 (Uniqueness). On two values the structureless laws are the identity and complement. The graph of the identity excludes nothing. Any structureless law that excludes something is complement.

L3 (Retorsion). If a tokening marks “no distinction is drawn” differently from its denial, a distinction is drawn. Discriminated tokening of the denial refutes the denial.

Th. 1. $\neg\text{Fix}(\neg)$: $!x \neq x$ on $\mathbb{B}$, and $X \leftrightarrow \neg X$ is unsatisfiable in Prop.

Th. 2. $\text{Run}(\neg)$: the orbit from each seed is the unique period-two alternation; the two orbits are relabelings of each other.

Th. 3. $\neg\text{Fix}(\neg) \leftrightarrow \text{Run}(\neg)$ (same instances; each direction is instantiation).

Th. 4. If a tokening marks the denial of occurrence differently from its assertion, occurrence holds and the two tokens are distinct. On two values, distinctness coincides with being one complement apart.

P. The token of the denial and the token of the assertion stand in oriented succession (forward holds; reverse fails). One instance per realized step; a full run requires infinitely many pairwise-distinct loci and discrimination at each step. P is independent of Th. 1–4 relative to the world signature (T, h, tok) : two tensed structures over one world agree on all claims and disagree on P .

Th. 5 (from P). Discrimination plus one oriented instance yields distinct loci, complementary successive values, and non-stillness. A chain of oriented instances realizes the run.

Th. 6. Claims are properties of succession-free worlds. If A is undeniable (true wherever its denial is discriminated) and entails running, then no still world discriminates the denial of A . Given a still discriminating world, no undeniable substitute for P entails running. Such a world is exhibited by ostension.

Th. 7. On loci with decidable equality, every relabeling-invariant succession is symmetric. Hence no invariant succession is oriented, and the orientation clause of P is not available from invariance.

Corollary. The alternation exists and is unique up to seed relabeling; nothing static answers to the ground (Th. 1–3). Discriminated denial witnesses occurrence (Th. 4). Given P , each such denial realizes a step (Th. 5). P is required for the temporal reading (Th. 6–7).

Commitments. (1) Verified conditionals (empty axiom footprint). (2) Five regimentations: ground as law; structureless as equivariant; groundhood as exclusion; satisfaction as objectual or processual; claims as properties of succession-free worlds. (3) One empirical input: a discriminating tokening, discharged by ostension; stepwise discrimination in a run is part of (4). (4) Premise P : one oriented instance per step; full run needs infinitely many loci and stepwise discrimination. (5) Constructive metalogic.

L1–L3 and Th. 1–4: `lean/Becoming.lean`. Th. 5–7, P , and independence: `lean/Obstruction.lean`. Both are Lean 4, prelude only, no imports, no axioms. Axiom audits: 52 and 47 lines respectively.